Business Math Drills

Core skills trio, part 3 of 3 · Numbers: how the argument is quantified on the spot. The other two: `presentation_and_storyline_guide.md` · `qa_drills.md`. Folder overview: `README.md`.

At Asper, phones and smartwatches are banned in the prep room, and practitioner judges trust teams that quantify on the spot. These 40 drills build speed on the calculations that appear in almost every case. House rules: round aggressively, state assumptions aloud, always give units, and sanity-check against an anchor. Speak every answer in English — "one point two billion Canadian dollars," never "1,2 Milliarden" — and remember the false friend: German *Billion* = English *trillion*; *Milliarde* = billion.

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Anchor numbers (memorize cold)

Anchor	Value	Anchor	Value
Germany population	84M	Canada population	40M
German households	41M (~2.0 per HH)	Canadian households	15M (~2.6 per HH)
Berlin	3.7M	Manitoba	1.4M
Munich	1.5M	Winnipeg (metro)	0.85M
EUR → CAD	≈ 1.5	Working days per year	250
German GDP	≈ EUR 4.3T	Canadian GDP	≈ CAD 2.9T

Fractions to percent, instantly: 1/6 ≈ 17%, 1/7 ≈ 14%, 1/8 = 12.5%, 1/9 ≈ 11%, 1/12 ≈ 8%.

Toolkit

- **Rule of 72:** years to double ≈ 72 ÷ growth rate. Rule of 114 for tripling.
- **Flip trick:** a% of b = b% of a (16% of 25 = 25% of 16 = 4).
- **Reciprocal shortcuts:** ÷0.8 = ×1.25; ÷0.9 ≈ ×1.11; ÷0.75 = ×1.33.
- **Successive changes multiply:** +30% then −30% = 1.3 × 0.7 = 0.91 → net −9%.
- **Compounding approximation:** n years at rate g ≈ n·g + [n(n−1)/2]·g² total growth.
- **Margin vs. markup:** margin = markup ÷ (1 + markup); a 25% markup is a 20% margin.
- **pp vs. %:** 20% → 25% is +5 percentage points, which is a 25% relative increase.

A. Market sizing (drills 1–8)

1. How many coffee shops does Berlin have? 3.7M people; assume one shop per 2,000 residents → ≈ 1,850, call it **2,000 shops**. Sanity: dense cafe culture, right order of magnitude.

2. Size the German cat-food market (EUR per year). 41M households × 25% with cats × 1.5 cats ≈ 15M cats; × EUR 300 per cat → ≈ **EUR 4.5B**. Bottom-up cross-check: 15M cats × ~EUR 0.55 of food per day ≈ **EUR 3B** — the two differ by ~40%, so quote the range and name the driver (spend per cat). Saying "between

three and four and a half billion, depending on whether you assume 55 cents or 80 cents a day" beats a single false-precise number.

3. Winter tires sold in Germany per year. ~49M passenger cars; a set of four replaced every 6 years → 49 ÷ 6 ≈ 8M sets ≈ 32M tires; × EUR 100 → ≈ **EUR 3.2B retail**.

4. Size the Canadian market for hockey skates. 40M × 12% who skate ≈ 5M skaters; replacement every 5 years → 1M pairs per year; × CAD 175 → ≈ **CAD 175M** (state the participation assumption — judges buy structure, not precision).

5. Annual candle spending in Manitoba (Boardroom-style local sizing). 1.4M ÷ 2.5 ≈ 560k households; 50% buy candles × CAD 40 per year → ≈ **CAD 11M**. The case insight hiding in the number: a premium local brand's home market is tiny, so growth must come from outside the province.

6. Gen Z banking prospects in Manitoba. 1.4M × 20% aged 13–28 ≈ 280k; credit unions plausibly reach a quarter → ≈ **70k prospects**; at CAD 150 revenue per young member → a **CAD 10M revenue pool**.

7. Bottled water in Germany. 84M × ~100 liters per capita × EUR 0.50 per liter → ≈ **EUR 4B**.

8. Public EV chargers needed in Germany by 2030. Assume 15M EVs; one public charger per 10 EVs → **1.5M chargers**; at EUR 10k average installed cost → an **EUR 15B infrastructure market**.

B. Margins and pricing (drills 9–15)

9. Revenue EUR 2.4M, COGS EUR 1.5M. Gross margin? → 0.9 ÷ 2.4 = **37.5%**.

10. Cost EUR 40, price EUR 50. → Margin 10/50 = **20%**; markup 10/40 = **25%**. Never confuse the two.

11. Cost CAD 91, target margin 35%. Price = 91 ÷ 0.65 = **CAD 140**.

12. 60% of revenue at 30% margin, 40% at 55%. Blended = 0.6(30) + 0.4(55) = **40%**.

13. Price EUR 50, contribution EUR 20. Cut price 10% → contribution EUR 15. Volume needed for the same profit: 20 ÷ 15 → **+33%**. (A 10% price cut demands a third more volume — say this in Q&A and watch judges nod.)

14. Fixed costs EUR 400k, contribution margin 25%, revenue EUR 2.0M → profit EUR 100k. Revenue +10% → contribution +EUR 50k → profit EUR 150k: **+50% profit from +10% revenue** (operating leverage of 5×).

15. Margin rises from 20% to 25%: **+5 pp, a 25% relative gain** — on EUR 10M revenue, +EUR 500k profit.

C. Growth and CAGR (drills 16–21)

16. At 12% growth, revenue doubles in 72 ÷ 12 = **6 years**.

17. 40 → 58 in 4 years: ratio 1.45; since $1.1^4 \approx 1.46$ → ≈ **10% CAGR** (precisely 9.7%).

18. A market of 8B growing 7% for 5 years: 35% simple + ~5% compounding ≈ +40% → ≈ **11.2B**.

19. 120 → 90 in 3 years: ratio 0.75; $0.91^3 \approx 0.75$ → ≈ **−9% CAGR**.

20. 6% for 8 years: 48% + (28 × 0.36%) ≈ **+58% total** (exact: +59%).

21. Revenue triples in 10 years: 114 ÷ 10 → ≈ **11–12% CAGR**.

D. Break-even (drills 22–27)

22. Fixed EUR 300k; price EUR 25, variable cost EUR 15 → 300k ÷ 10 = **30,000 units**.

23. Fixed EUR 180k; contribution margin 30% → break-even revenue 180 ÷ 0.30 = **EUR 600k**.

24. A machine costing CAD 240k saves CAD 8 per unit → pays for itself at **30,000 units**.

25. Fixed EUR 200k; price falls EUR 20 → EUR 18, variable cost EUR 10: break-even moves from 20k to 25k units — **a 10% price cut raises break-even by 25%**.

26. Two products in a 50/50 unit mix, contributions EUR 8 and EUR 16 → weighted EUR 12; fixed EUR 600k → **50,000 units**.

27. Launch cost CAD 90k; monthly contribution CAD 7.5k → **12 months to break even**.

E. Unit economics (drills 28–33)

28. CAC EUR 60; order value EUR 45 at 40% margin → EUR 18 per order → 60 ÷ 18 = 3.3: **the customer turns profitable on the 4th order**.

29. Monthly churn 4% → average lifetime ≈ 1 ÷ 0.04 = 25 months; ARPU EUR 12 × 70% margin × 25 = **LTV EUR 210**; CAC EUR 70 → **LTV/CAC = 3.0** (healthy at ≥ 3).

30. Subscription CAD 10; COGS 2 + support 1 → 70% margin → CAD 7 per month; CAC 42 → **6-month CAC payback**.

31. Delivery order: basket CAD 30 × 25% take rate = CAD 7.50 revenue; delivery 5.00 + support 1.00 → **CAD 1.50 profit per order (5% of basket)**.

32. 500 customers × CAD 200 per month = CAD 100k MRR → **CAD 1.2M ARR**.

33. One store: sales EUR 1.2M; COGS 55% (660k), staff 250k, rent 120k, other 80k → **EBITDA EUR 90k = 7.5% margin**.

F. Quick NPV and payback (drills 34–37)

34. Cash flow EUR 150k per year forever at 10% → perpetuity value 150 ÷ 0.10 = EUR 1.5M; investment EUR 1.0M → **NPV +EUR 0.5M**.

35. Invest 600k; cash flows 120 / 180 / 240 / 300: cumulative 540 by Year 3; need 60 of Year 4's 300 → **payback ≈ 3.2 years**.

36. Annuity factor, 5 years at 10% ≈ 3.8; cash flow 200k → PV 760k − 700k invested = **NPV ≈ +60k**. Memorize the 10% factors: 3 years ≈ 2.5, 5 years ≈ 3.8, 10 years ≈ 6.1.

37. Growing perpetuity: 100k at r = 12%, g = 2% → 100 ÷ (0.12 − 0.02) = **EUR 1.0M**.

G. Mental-math finishers (drills 38–40)

38. 16% of 25 → flip it: 25% of 16 = **4**.

39. Cost 96, target 20% margin → 96 ÷ 0.8 = 96 × 1.25 = **120**.

40. A campaign lifts sales 30%, then a stockout cuts 30% → 1.3 × 0.7 = 0.91 → **net −9%**. Sequential percentages never cancel.

The weekly 30-minute drill routine

Minutes	Activity
0–5	**Anchor recall.** A partner quizzes populations, households, FX, and the fraction-percent table — answers spoken in English, with units.
5–15	**Two market sizings aloud** — one German, one Canadian context, 5 minutes each. The partner must challenge one assumption; the sizer defends it or adjusts on the spot.
15–25	**Timed gauntlet.** Eight mixed drills from sections B–G, 75 seconds each, no calculator; answers written, then checked.
25–30	**Error log.** Every miss is re-derived and recorded; the log becomes next week's warm-up.

Rotate the quizmaster weekly. Once a month, run a full mock Q&A in which the Chief Analyst defends a case model against hostile numbers questions from the coach — drills only count once they survive contact with a judge.

The standard for every answer, every time: a rounded number, its unit, the assumption that produced it, and one sanity check. "Roughly CAD 11 million per year, assuming half of Manitoba's 560,000 households spend 40 dollars — small enough that growth must come from outside the province." That sentence is the drill.